D 92218 (Pages : 3) Name.....

Reg. No.....

# THIRD SEMESTER B.Com./B.B.A. DEGREE EXAMINATION NOVEMBER 2015

(CUCBCSS—UG)

Common Course

A 11—BASIC NUMERICAL SKILLS

Time : Three Hours Maximum : 80 Marks

## Part A

*Answer* **all** *questions.*
*Each question carries* 1 *mark.*

I. Choose the correct answer :

1 The middle most item of a distribution is :

(a) Mean. (b) Mode.

(c) Median. (d) None of these.

2 Reciprocal of X values is related with :

(a) Harmonic Mean. (c) Linear equation.

(b) Mean deviation. (d) All of the above.

3 Which of the following is a measure of central tendency ?

(a) Standard deviation. (b) Range.

(c) Median. (d) Kurtosis.

4 The median of 35, 23, 45, 50, 80, 61, 92, 40, 52, 61 :

(a) 51. (b) 80.

(c) 50. (d) 40.

5 The compound interest for Rs. 7,000 for 4 years ,interest payable half yearly at 6 % is :

(a) 1,867. (b) 1,000.

(c) 300. (d) 2,000.

II. Fill in the blanks :—

6 A set which contains no elements is ————.

7 ———— ignores + or – sign while computing measure of dispersion.

Turn over

8 If A = {1, 2, 3} B = {2, 5,} then A ∪ B is ———.

9 ——— is considered as the ideal index number.

10 The second term of G P series is 9 and the fifth term is 243, the fourth term of series is ———.

(10 × 1 = 10 marks)

## Part B (Short Answer Questions)

*Answer any* **eight** *questions.*
*Each question carries* 2 *marks.*

11 The sum of three continuous terms in G P is 35 and their product is 1000. Find the terms.

12 What do you mean by harmonic Mean ?

13 What is an index number ?

14 If U = {1, 2, 3, 4, 5} and A = {2, 4, 5}, Find $A^c$.

15 If $A = \begin{bmatrix} 1 & -3 & 2 \\ 0 & 3 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 5 & 6 \\ 1 & 1 & -2 \end{bmatrix}$ find 2A – 3B.

16 Represent, A ∪ B A ? B, and A – B by means of venn diagram.

17 Solve $5x^2 - 125 = 0$.

18 Of the 100 borders of a hostel 80 drink tea , 40 drink coffee and 20 drink tea and coffee. How many of them drink neither tea nor coffee ?

19 Explain the limitations of statistics.

20 Explain the uses of index numbers.

(8 × 2 = 16 marks)

## Part C

*Answer any* **six** *questions.*
*Each question carries* 4 *marks.*

21 Find an appropriate index from the following data :—

| Items | Price 2004 | Price 2007 | Quantity consumed |
|---|---|---|---|
| A | 2 | 4 | 3 |
| B | 8 | 9 | 5 |
| C | 7 | 8 | 8 |
| D | 5 | 6 | 2 |

22 If $A\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$ and $B\begin{bmatrix}5 & 6\\ 0 & -2\end{bmatrix}$ verify whether AB = BA.

23 Draw a histogram to the frequency distribution showing the ages of people :

| Ages | : | 10 – 15 | 15 – 20 | 20 – 30 | 30 – 40 | 40 – 50 | 50 – 75 | 75 – 100 |
|---|---|---|---|---|---|---|---|---|
| Frequency | : | 4 | 12 | 20 | 18 | 14 | 25 | 10 |

24 By means of Venn diagram, prove that $(A \boxed{?} B) = A' \cup B'$ (Demorgan's law).

25 Find the standard deviation and co-efficient of variation of the values

10, 12, 80, 70, 60, 100, 0, 4.

26 What are the main types bar diagram ?

27 The population of a country increases every year by 2.4 % of the population at the beginning of that year. In what time will the population double itself ? Answer to the nearest year.

28 What are the functions of statistics

(6 × 4 = 24 marks)

## Part D (Essay Questions)

*Answer any* **two** *questions.*

*Each question carries* 15 *marks.*

29 With median as the base calculate the mean deviation and compare the variability of the two series A and B :

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Series A : | 3484 | 4572 | 4124 | 3682 | 5624 | 4388 | 3680 | 4308 |
| Series B : | 487 | 508 | 620 | 382 | 408 | 266 | 186 | 218 |

30 Find the two numbers whose sum is 18 and product is 72.

31 Explain in detail the steps in statistical investigation.

(2 × 15 = 30 marks)